PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. Under the Panjab University Act VII of 1947 –enacted by the Govt. of India)

SYLLABI

Course Outline for
B.Voc. (Food science and Technology)

as per NEP 2020

w.e.f. session 2026-27

Course Outline for B.Voc. Food science and Technology
as per NEP 2020 w.e.f. session 2025-26

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FST 101	Core Discipline	Introduction To Food Microbiology	3	1	4	40	10	40	10	100
FST 102	Core Discipline	Bakery And Confectionery Technology-I	3	1	4	40	10	40	10	100
FST 103	Core Discipline	Bakery And Confectionery Technology-II	3	1	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	Common Basket			2					50
SEC I	Skill Enhancement Course 1	Common Basket	-	-	3					75
IDC I	Interdisciplinary Course MDSC 1	Common Basket	-	-	3					75
VAC I	Value-Added Course1	Common Basket	-	-	2					50
PW I	Project Work	Project Work-I& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FST 201	Core Discipline	Dairy Technology	3	1	4	40	10	40	10	100
FST 202	Core Discipline	Milk And Dairy Food Products	3	1	4	40	10	40	10	100
FST 203	Core Discipline	Management of Food Industry	4	-	4	80	20			100
AEC / MIL	Ability Enhancement Course	Common Basket	-	-	2					50
SEC II	Skill Enhancement Course 2	Common Basket	-	-	3					75
IDC II	Interdisciplinary Course MDSC2	Common Basket MDSC2	-	-	3					75
SIT 201*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role –Baking Technician, Dairy Products Processor

*Summer Industrial Training of 2-4 weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 3rd Semester. Viva-Voce examination to be held within 3 weeks of the start of 3rd semester.

** Minimum Credits required to get Certificate Course in Food Science and Technology = 48 + 4 #.

#Exit 1: Certificate Course-Student can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

Semester-III

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FST 301	Core Discipline	Food Microbiology	3	1	4	40	10	40	10	100
FST 302	Core Discipline	Food Analysis and Instrumentation	3	1	4	40	10	40	10	100
FST 303	Core Discipline	Microbiological Analysis	3	1	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	Common Basket	-	-	2					50
SEC I	Skill Enhancement Course 3	Common Basket	-	-	3					75
IDC III	Interdisciplinary Course MDSC 3	Common Basket	-	-	3					75
VAC I	Value-Added Course2	Common Basket	-	-	2					50
PW II	Project Work	Project Work-I& Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-IV

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FST 401	Core Discipline	Food Quality Testing and Evaluation	4		4	80	20			100
FST 402	Core Discipline	Technology of cereals, pulses and oil seeds	3	1	4	40	10	40	10	100
FST 403	Core Discipline	Food Safety	3	1	4	40	10	40	10	100
FST 404	Core Discipline	Food Quality Management	3	1	4	40	10	40	10	100
AEC / MIL	Ability Enhancement Course	Common Basket	-	-	2					50
VAC III	Value-Added Course3	Common Basket	-	-	2					75
SIT II*	Internship	Summer Industrial Training (SIT)*	-	4	4	-	-	100	-	100
Total Credits					**24**					**600**

Job Role – Food Microbiologist

*Summer Industrial Training of 2-4weeks in a relevant Industry after 4thSemester Examinations during summer break. Training report by the student to be submitted within one week of start of 5thSemester. Viva-Voce examination to be held within 3 weeks of the start of 5th semester.

Semester-V

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FST 501	Core Discipline	Principles of Management	4	-	4	80	20			100
FST 502	Core Discipline	Production Planning In Food Industry	3	1	4	40	10	40	10	100
FST 503	Core Discipline	Production Optimization & Cost Efficiency	3	1	4	40	10	40	10	100
FST 504	Core Discipline	Food Plant Layout and Business Ethics	4	-	4	80	20			100
FST 505	Core Discipline	Food Biochemistry	4	-	4	80	20			100
VAC 4	Value-Added Course	Common Basket	-	-	2					50
PW II	Project Work	Project Work-I& Viva-Voce	-	-	2					50
Total Credits					**24**					**600**

Semester-VI

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
FST 601	Core Discipline	Budgetary control and optimization	4	-	4	80	20			100
FST 602	Core Discipline	Documentation system management	3	1	4	40	10	40	10	100
FST 603	Core Discipline	Food industry safety and environmental management	3	1	4	40	10	40	10	100
FST 604	Core Discipline	Food Packaging	3	1	4	40	10	40	10	100
FST 605	Core Discipline	Applied Nutrition	4	-	4	80	20			100
FST 606	Core Discipline	Food Waste Management and Sustainability	3	1	4	40	10	40	10	100
Total Credits					**24**					**600**

Project Work to be undertaken after 5th Semester Examinations during winter break. Project report by the student to be submitted within one week of start of 6th Semester. Viva-Voce examination to be held within 3-weeks of the start of 6th semester.

ESSENTIAL COMPONENTS OF COURSE OF B.Voc. Food science and Technology (SEMESTERS I & IV)

PROGRAMMEOUTCOMES

PO1. They will be able to demonstrate knowledge of Food science, Food Microbiology, and Food Technology.

PO3. They will demonstrate an ability to identify, formulate and solve processing problems related to food industry

PO4. They will able to focus on the importance of safe processed nutritious food.

PO5. They will get exposure to various technologies aspect of Food product preparation.

PO6. They will demonstrate an ability to design or process food products as per the needs and specifications.

PO7. They will demonstrate an ability to work in Food industries, research organization and teaching.

Semester I

1	**Course Title**	**Introduction To Food Microbiology**
2	**Course Code**	**FST 101**
3	**Typology of Course**	**Core Discipline:**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><th>External Theory</th><th>Internal Theory</th><th>External practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table> a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Baking Technician**
8	**Course Objectives**	**To understand the types of food microbes, causes for food spoilage, process for food spoilage, need for food preservation.**
9	**Course Outcomes**	On successful completion of the course, the Students will be able **CO: To get a basic knowledge about significance of microbes in different types of food, microbial growth characteristics, various methods of food preservation, hurdle technology etc.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions,PPT,Practical and Group Discussions, Case
11	**Instructions for paper setters and candidates**	1. **The syllabus of this paper has been divided into FOUR units.** 2. **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** 3. **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** 4. **All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

Unit I

Introduction to Food Microbiology
• Definition and Scope of food microbiology
• Characteristics of Microorganisms in Food
• Types of microorganisms, Classification and Nomenclature, Morphology and Structure and their importance in food (bacteria, fungi, viruses and prions, protozoan and others)
• Significance of spores
• Heat resistance of microorganisms and their spores –bacteria and their spores – fungi and their spores – yeast and their spores - spoilage of canned foods and types of spoiled cans

Unit II

• Microbial Growth in Food
• Microbial Growth Characteristics- Bacterial growth curve, microbial reproduction and microbial growth in food
• Factors affecting the growth of microorganisms in food

Unit III

Microbial Food Spoilage
• Sources of Microorganisms in foods
• Some important food spoilage bacteria
• Food spoilage: Microbial, physical, chemical & miscellaneous.
• Changes caused by micro-organisms during spoilage (breakdown of proteins, carbohydrates, fats and other constituents)
• Detection of food pathogens: Overview of Conventional and Rapid methods to detect food pathogens.

Unit IV

Control of Microorganisms in Foods
• Principles and methods of preservation
• Physical Methods of Food Preservation- Dehydration, Freezing, Cool Storage, Heat Treatment (esp. thermo bacteriology), Irradiation
• Chemical Preservatives
• Bio preservatives
• New Non Thermal methods
• Introduction to Hurdle concept and Predictive Microbiology

Practical

Credits :1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. Introduction to the Basic Microbiology Laboratory Practices and Equipments
2. Functioning and use of compound microscope
3. Cleaning and sterilization of glassware
4. Preparation and sterilization of nutrient broth
5. Cultivation and sub-culturing of microbes
6. Preparation of slant, stab and plates using nutrient agar
7. Morphological study of bacteria and fungi using permanent slides
8. Methods of sterilization and preparation of media
9. Simple staining
10. Endospore staining
11. Standard Plate Count Method

13 Suggested Resources for Reading

1) Frazier William C and Westhoff, Dennis C. Food Microbiology, TMH, New Delhi, 2004

2) Jay, James M. Modern Food Microbiology, CBS Publication, New Delhi, 2000

3) Garbutt, John. Essentials of Food Microbiology, Arnold, London, 1997

4) Pelczar MJ, Chan E.C.S and Krieg, Noel R. Microbiology, 5th Ed., TMH, New Delhi, 199

5) Adams M.R and Moss M.O, 2004 "Food Microbiology", Panima Publishing corporation, New Delhi, 2ndEdition.

6) Sivasankar B, 2009. "Food Processing and Preservation", PHI Learning Private Limited, Eastern Economy Edition, 6th edition,

7) William C Frazier and Dennis C. Westoff , 2008. "Food Microbiology", Special Edition, Springer, The Mc Graw-Hill Companies.

Note: Latest edition of text books may be used.

1	**Course Title**	**Bakery And Confectionery Technology-I**
2	**Course Code**	**FST102**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30hours of Practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	**External Theory: 40 | Internal Theory: 10 | External practical: 40 | Internal Practical: 10** a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Baking Technician**
8	**Course Objectives**	• To provide knowledge about the machinery and process involved in the baking process • To understand the various types of sugar and its grades • To know the confectionery product manufacture
9	**Course Outcomes**	On successful completion of the course, the Students will be able to CO1. Demonstrate a comprehensive understanding of the machinery, processes, and techniques involved in bakery and confectionery production, encompassing ingredient selection, mixing methods, baking temperatures, and quality control measures. CO2. Develop technical skills in operating bakery and confectionery equipment, conducting quality tests on raw materials and finished products, and implementing hygiene and sanitation practices to ensure food safety and product quality
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

Unit I

- Introduction to food processing and preservation.
- Various products of the bread and bakery sub-sector
- List the various types of industries within the bakery sub-sector
- Explain the baking process
- Equipments used in Bakery (Dough mixers, Dividers, rounders, Proofing, molding, Ovens, Slicers etc.), Cleaning and maintenance of the work area and machineries, Baking ingredients required for production and plan production sequence
- Testing of Flour for Bakery Goods: Laboratory testing of Wheat grain Quality, Moisture tests, Grain hardness testing. Testing, Visco graph, Amylograph, Ferinograph. Units of measurement used in the baking industry.
- Raw material required for bakery products. Role of flour, water, yeast, salt, Sugar milk and fats.
- Yeast- An elementary knowledge of Baker's yeast, the part it plays in th fermentation of dough and conditions influencing it's working.
- Effect of over and under fermentation and under proofing of dough and other fermented goods.

Unit II

- Process of mixing and knead ingredients to make dough
- Oven & Baking – Knowledge and working of various types of oven.
- Biscuit-Types of biscuit dough – Developed dough, short dough, semi-sweet, enzyme modified dough and batters- importance of the consistency of the dough, Factors affecting the quality of biscuits/ cookies
- Cake – Flour specification – ingredients: Cake making ingredients- flour, sugar, shortening and egg types of chemically aerated goods Moistening agents Fats and Oil leavening agents
- Manufacturing process – Cake making methods sugar batter process, flour batter process, genoise method and blending and rubbing method, Correct temperature for baking different varieties of cakes, Balancing of cake formula Characteristics of cakes: external: Internal ,Types of icing.

Unit III

- Types of Confectionery, raw materials and processing of chocolates, hard boiled candies.
- Additives for Confectioneries.
- Equipments used in Confectionery manufacture
- Toffees – Milk toffee; chocolate; stick jaws; liquor chocolate.
- Baking temperatures for confectionery goods.
- Process of proofing process of baking products in the oven
- Process of cooling of baked products.

Unit IV

Bread manufacturing process – Straight dough fermentation Bread improvers- improving physical quality Methods of bread making

- Straight dough methods
- Delayed salt method
- No time dough method
- Sponge and dough method Characteristics of good bread
- External Characteristics – volume, symmetry of shape
- Internal characteristics- color, texture, aroma, clarity and elasticity Bread faults and their remedies.
- Bread diseases- Rope and Mould – causes and prevention.

Practical

Credits:1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. **Preparation of cookies and biscuits**
2. **Preparation of different types of cakes**
3. **Preparation of different types of pastries**
4. **Preparation of different types of ice creams (Vanilla, Strawberry, chocolate, Pineapple, Mango)**
5. **Preparation different types of Pudding**
6. **Preparation of Bread rolls; Bread sticks & softs rolls;**
7. **Preparation of Fermented doughnuts;**
8. **10. Preparation of Pizza base**

13 Suggested Resources for Reading

1. Samuel A. Matz, "Bakery Technology and Engineering", Chapman & Hall, 3rd Edition, 1992.

2. Reference Books 1. "Association of Operative Millers Cereal Millers Hand Book", Burgess Publishing company, USA, 1963.

3. Pomeranz Y, "Modern Cereal science and Technology" MVCH Publications, NY, 1987.

4. Kent N.L.,.Evers A.D, "Technology of Cereals" Peregaman Press, Elsevier Publishers, 1994.

5. Samuel A. Matz, "Equipment for Bakers" Pan Tech International Publication, 1988.

6. Stanley P Cauvain, Linda S Young, "Technology of Bread making", Aspen publication, 2ndEdition, 2007.

7. "Association of Operative Millers Cereal Millers Hand Book", Burgess Publishing company, USA, 1963.

8. Kent N.L.,.Evers A.D, "Technology of Cereals" Peregaman Press, Elsevier Publishers, 1994.

Note: Latest edition of text books may be used.

<table>
<tr><td>1</td><td>Course Title</td><td>Bakery And Confectionery Technology-II</td></tr>
<tr><td>2</td><td>Course Code</td><td>FST 103</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>45 hours of lectures for Theory + 30 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
3-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Baking Technician</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To know the manufacturing of Bakery and confectionery product</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able
CO1. To acquire problem-solving abilities to tackle challenges related to ingredient sourcing, production planning, equipment maintenance, and quality assurance in bakery and confectionery operations.
CO2. To demonstrate professional responsibility, adhere to industry standards, regulations, and best practices, including compliance with food safety laws, hygiene standards, and waste management protocols.
CO3.To provide students’ knowledge about the machinery and process involved in the baking process</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions,PPT,Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

Unit I

- Bakery layout- The required approvals for setting up of a Bakery- Government procedure and Bye laws.
- Selection of site & Selection of equipment
- Layout design
- Electricity
- Process of cleaning the work area and machineries after production
- Roles and responsibilities of a bakery production specialist.
- Plan, organize and management of work in bakery.
- Method of documenting and recording the details of baking ingredients to final finished product

Unit II

- Quality control of raw material and of finished products
- Role of Hygiene in Bakery
- Personal hygiene, care of skin, hand and feet, Food handlers hygiene, protective clothing.
- Dishwashing methods- manual and machine dish washing- merits and demerits
- Garbage disposal- different methods- advantages and disadvantages
- Food poisoning- Causative factors and the precautions to be taken by food handlers.

Unit III

- Food Storage- Techniques of correct storage, storage temperature of different commodities prevent bacterial manifestation or contamination. Storage of confectionery products.
- Pest Control- Rodents and insect control techniques, special stress on control of flies , rats and cockroaches, care of the premises and equipment
- Municipal health laws.
- Golden rules of first aid and treatment for cuts, wounds, burns
- Process of packaging of baked products

Unit IV

- Accounts and costing:
- Purchase book, purchase return book, Stores requisition.
- Sales book, Sales return book, Cash voucher. Credit voucher book
- Preparation of invoices and debit/ credit memos
- Percentage and discounts
- Introduction to ingredient costs, labour costs, overheads, gross profit calculation of cost price, sales price and catalogue price.
- Commodities: Sugar, Cocoa, Chocolate, Milk, Butter, Cream, Cheese, Food colours, Flavors & essences
- Dry fruits & nuts used in confectionery, Fresh and preserved fruit products
- Food laws- Agmark

Practical

Credit :1
Practical: 40
Internal Assessment: 10
Total Marks: 50

- **Prepare the Bread rolls/cakes/ and assessment of its quality**
- **Perform the quality test on fat/butter for bakery applications on different parameters**
- **Carry out the assessment of market Candy and Chewing gum.**
- **Perform the quality assessment test on yeast and skimmed milk powder for bakery application.**
- **Prepare Chocolate cookies and assessment of its quality.**

13 Suggested Resources for Reading

1. Samuel A. Matz, "Bakery Technology and Engineering", Chapman & Hall, 3rd Edition, 1992.

2. Reference Books 1. "Association of Operative Millers Cereal Millers Hand Book", Burgess Publishing company, USA, 1963.

3. Pomeranz Y, "Modern Cereal science and Technology" MVCH Publications, NY, 1987.

4. Kent N.L.,.Evers A.D, "Technology of Cereals" Peregaman Press, Elsevier Publishers, 1994.

5. Samuel A. Matz, "Equipment for Bakers" Pan Tech International Publication, 1988.

6. Stanley P Cauvain, Linda S Young, "Technology of Bread making", Aspen publication, 2ndEdition, 2007.

7. "Association of Operative Millers Cereal Millers Hand Book", Burgess Publishing company, USA, 1963.

8. Kent N.L.,.Evers A.D, "Technology of Cereals" Peregaman Press, Elsevier Publishers, 1994.

Note: Latest edition of text books may be used.

Semester 2

1	**Course Title**	**DAIRY TECHNOLOGY**
2	**Course Code**	**201**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	**External Theory: 40** | **Internal Theory: 10** | **External practical: 40** | **Internal Practical: 10** a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Dairy Products Processor
8	**Course Objectives**	**To know the compositional and technological aspects of milk** **To study processed milk products**
9	**Course Outcomes**	**On successful completion of the course, the Students will be able** **CO1. To understand the technology and constituents of dairy products and their preparation** **CO2. To develop technical proficiency in various areas such as milk composition analysis, dairy processing techniques, quality control measures, and sanitation protocols in dairy industry settings. Additionally, students will acquire problem-solving skills to identify, analyze, and address challenges related to milk processing, product development, quality assurance, and food safety in dairy operations.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions,PPT,Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**
12	**Course Content**	**Credits: 3** **Theory: 40** **Internal Assessment: 10** **Total Marks: 50**

Unit I

TECHNOLOGY OF MILK AND MILK PRODUCTS

Composition and Physical properties of milk , milk color, taste, pH and buffering capacity, refractive index, viscosity, surface tension, freezing, boiling point, specific heat, OR, electrical conductivity.

Unit II

Milk fat

• Composition and structure, factors affecting melting point, boiling point, solubility and Refractive Index, fat constants (saponification value, iodine value, RM value, Polenske value, peroxide value).

• Chemical reactions of fat (hydrolysis, auto-oxidation), condition favoringauto oxidation, prevention, measurement of auto-oxidation.

Unit III

Fats and Oils

• Classification of lipids,

• Rancidity - hydrolytic and oxidative rancidity and its prevention.

• Define - margarine, butter, hydrogenated vegetable oil, lard.

Protein and Enzymes

• General structure, amphoteric nature, difference between casein and serum protein, different types of casein (acid and rennet), uses of casein.

• Enzymes- catalase, alkaline phosphatase, lipases and proteases.

UNIT IV

• Liquid milk handling operations: Systems of collection of milk, Reception, Platform testing, Process of testing milk for accepted quality standards, organoleptic test of milk, Procedure for COB test of milk

• Various stages of processing: Filtration and Clarification, Skimming, Standardization Homogenization, Pasteurization (LTHT, HTST, UHT) Sterilization, Packaging, Storage and distribution of fluid milk

• Description and working of clarifier, cream separator, homogenizer and plate heat exchanger. Need for processing milk.

• List the various units within a dairy processing plant

• List the machineries used in a dairy processing plant

• Roles and responsibilities of a dairy products processor in a dairy processing plant

PRACTICAL

Credits: 1
Practical: 40
Internal Assessment: 10
Total Marks: 50
Time: 3 hours

1. To perform platform tests in milk.(Acidity, COB, MBRT, specific gravity, SNF)
2. To estimate milk protein by Folin method.
3. To estimate milk fat by Gerber method.
4. Preparation of flavored milk/. Pasteurization of milk
5. To prepare casein and calculate its yield.

13 Suggested Resources for Reading

1. De Sukumar, Outlines of Dairy Technology, Oxford University Press, Oxford. 2007
2. Hall GM, Fish Processing Technology, VCH Publishers Inc., NY, 1992
3. Sen DP, Advances in Fish Processing Technology, Allied Publishers Pvt. Limited 2005
4. Shahidi F and Botta JR, Seafoods: Chemistry, Processing, Technology and Quality, Blackie Academic & Professional,London,1994
5. Webb and Johnson, Fundamentals of Dairy Chemistry

Note: Latest edition of text books may be used.

1	**Course Title**	**MILK AND DAIRY FOOD PRODUCTS DEVELOPMENT**
2	**Course Code**	**202**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	**External Theory: 40** | **Internal Theory: 10** | **External practical: 40** | **Internal Practical: 10** a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Dairy Products Processor
8	**Course Objectives**	**To enable the students to understand the various dairy products and their preparation.**
9	**Course Outcomes**	On successful completion of the course, the Students will be able CO1. To apply theoretical knowledge and practical skills acquired during the course to optimize milk handling operations, improve product quality, and enhance efficiency in dairy processing plants, explore new concepts, technologies, and processes in dairy industry advancements, ethical conduct in dairy industry practices, including adherence to hygiene, sanitation, safety guidelines, and responsible waste management. CO2. To make the students aware about the diverse milk products and practically capable to develop new products for the market CO3. To demonstrate a comprehensive understanding of the compositional and technological aspects of milk, dairy products, and their development processes.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions,PPT,Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

Unit I

• Dairy industry in India & its future prospects.
• Process of production planning
• State the factors affecting operation efficiency during production
• New Concept in milk processing- UHT, Membrane processing, Microwave and irradiation treatments; aseptic packaging. Microbiology of milk, sources of milk contamination.

Unit II

• Process for producing dairy products and process of pasteurization :
• Ingredients required for production
• State the production process of pasteurization
• Process of separation and bactofugation
• Method of standardization of milk
• Method of homogenization of milk
• Method of heat exchange during pasteurization
• Process of HTST pasteurization
• Process of producing lassi, flavoured drinks, cheese dahi, kalakand, ice-cream, cream, butter , cooking butter, ghee

Unit III

• Technology of condensed and evaporated milk: Composition, nutritive value, process of manufacture, defects (their causes and prevention).
• Technology of yoghurt, Acidophilus milk, Bufgaricus milk, Kumiss, Kefir; Manufacturing of Cheddar, Mozzarella, cottage and processed cheese.
• Milk adulteration and quality control in dairy industry. By products of dairy industry and their utilization.
• Milk based infant foods; Manufacturing of casein, Caseinate, Co-precipitates, WPC, Lactose; National and International Organizations in dairy Industry.
• Milk and milk product standards and legislations in India: Grading of milk and criterion of grading, reconstituted milk, synthetic milk.
• List the different packaging materials used to pack dairy products
• Method of packaging dairy products

Unit IV

• Methods of cleaning and sanitization: Cleaning of production area, equipment, and tools used Equipment, detergents and sanitizers used in the cleaning and maintenance of the work area, Properties of the cleaning agents used, CIP method of cleaning,
• Lubrication system and its process followed in the dairy industry
• State the different types of maintenance procedures, Periodic maintenance of all production machineries
• Method of managing and disposing waste material
• Personal hygiene and sanitation guidelines
• Demonstrate how to use tools safely
• Food safety hygiene and quality standards to follow in a work environment , HACCP principles to eliminate food safety hazards in the process and products
• Method of documenting and recording the details of raw material to final finished product

PRACTICAL

Credits: 1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1 Platform tests,
2 Determination of fat, SNF, TS Protein,
3 Lactose and ash contents of milk.
4 Lay out plan for setting up of milk plant,
5 Preparation and Evaluation of different types of milk and milk products 6 Preparation and evaluation of butter, ice cream, cheese, yoghurt

13 Suggested Resources for Reading

1. Sukumar. 1983. Outlines of Dairy Technology. De,Oxford University Press.
2. James N.Warner.1989. Principles of Dairy Processing by, Wiley Eastern Ltd.
3. Eckles,Combs; and Macy. 1982. Milk and Milk Products by,Tata McGraw Hill.
4. Aneja et al. 2002.Technology of Indian Milk Products.A Dairy India Publication.

Note: Latest edition of text books may be used.

1	**Course Title**	**MANAGEMENT OF FOOD INDUSTRY**
2	**Course Code**	**FST 203**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory +00 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	**External Theory: 80** | **Internal Theory: 20** a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Dairy Products Processor**
8	**Course Objectives**	**To develop a knowledge base in key areas of institutional food management.** **To impart necessary expertise to run a food service unit.** **To provide practical level experience in managing food service management.** **To critically evaluate the functioning of food service units.**
9	**Course Outcomes**	On successful completion of the course, the Students will be able CO1. To impart necessary expertise to run a food service unit. CO2. To critically evaluates the functioning of food service units. CO3 To provide practical level experience in managing food service management
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

Credits: 4
Theory: 80
Internal Assessment: 20
Total Marks: 100

• Institutional management, Levels of Management.
• Review of different types of food service operations. Commercial, non- commercial food service institutions.
• Importance of food standards. Need for knowledge as to what constitutes good products, criteria for good quality products, hazards of poor quality, quality control, Specification for quality in food products.
• Organization - definition and types

Unit-II

•The Manufacturing (Production) Function: Objectives of Production Management, Operation Concept, Meaning of Product Development, Factors Influencing Choice of Manufacturing Systems, Classification of Manufacturing Systems;
• Basic Procedure of Method Study (Work Study) and Time Study; Concepts, Objectives and functions of Production Planning and Control (PPC).
• Physical plant- its location, floor plans, space allowance, Institutional food units-storage, preparation and serving
Cost control
a) Food cost, Labour cost, Maintenance cost
b) Budgets
c) Documentation and Records: Documentation and maintenance of record of raw materials and packaging materials
d) Portion control
• Role of supervisor in fruits and vegetable processing industry. • Planning and organization of work: organization standards, process standards and procedures followed in the organization, types of products produced by the organization, Code of business conduct, Dress code.

Unit-III

• Personnel Management
• Personnel and leadership qualities for food service administration.
• Labour , Types of labour, criteria for selection and employees training.
• Labour laws and legal aspects- health & safety of employees, welfare policies.
• Manage and lead a team Management Skills-The Importance of Delegation, Developing Your Team, Motivating Your Team: Motivation Theories – Maslow's, Herzberg's; Worker's Participation in Management; Collective Bargaining.
• Communicating and Working With Your Team – and With Others, Communicating With People Outside Your Team
• Managing Discipline

Unit-IV

Hygiene and Sanitation
i) Personal hygiene- Importance of personal cleanliness in handling and serving food. Health examination of personnel. ii) Sanitation in handling food and equipment.
a. Preventing contamination of cooked food and handling of fresh foods,. b. Cleaning and hygienic handling of equipments. c. Insect and Rodent control d. Safety, general safety rules in food preparation and service area. Accident prevention
Equipment- Types of equipment- criteria for selection, operation and care.
Food laws and regulations on raw materials, product(s), packaging and labelling
Food safety, hygiene, sanitation GMP , HACCP and quality assurance procedures
Procedures for disposal of waste.

13 Suggested Resources for Reading

1 West Wood A; Harper Food Service in Institution.
2. West, Bessin, Brooks; Food Service in Institutions
3. A.M. Home Economics Association; Hand Book of Food Preparations:
4. Sweetman, M.M. 4, Mac, Keller; Food Selection and Preparation:
5. Oliver B., Watson;School Lunch Room Service
6. Lender H. Katshever and Margret E. Terrel; Food Service Planning: Layout Equipment
7. Davidson and Passmore- Human Nutrition and Dietetics.

Note: Latest edition of text books may be used.

1	**Course Title**	**Food Microbiology**
2	**Course Code**	**301**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><th>External Theory</th><th>Internal Theory</th><th>External practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table> • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Food Microbiologist**
8	**Course Objectives**	**To introduce to students fundamental concepts of microbiology**
9	**Course Outcomes**	On successful completion of the course, the Students will be able CO. **To understand the fundamental concepts of microbiology that includes microbial diversity, their morphological and functional properties** **CO2: To understand the techniques to study microorganisms, growth and control of microorganisms as well as food spoilage and its control.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	1. **The syllabus of this paper has been divided into FOUR units.** 2. **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** 3. **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** 4. **All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

UNIT I

- **Introduction – Discovery of microbial world, Importance of microbiology, Microbial classification and nomenclature, Characteristics of major groups of microorganisms: Archaebacteria, Eubacteria, Fungi, Protozoa and Viruses and Bacteriophages.**
- **Principles of food microbiology and microbiology terminologies**
- **Significance of micro-organisms in preservation, production and spoilage of food**
- **Microbial Nutrition: Nutritional requirements of microbes; Types of culture media; Classification of microbes on the basis of nutritional requirements, Identification of bacteria.**

UNIT II

- **Human-Microbial Interactions: Normal flora –Gastrointestinal tract; Pathogenic mechanisms of food borne bacteria, Brief account of mechanisms of action of chemotherapeutic agents, Introduction to specific and nonspecific defense mechanisms to infections.**
- **Enumeration of Microorganisms- qualitative and quantitative**
- **Food Fermentations**
- **Fermentation –definition and types**
- **Microorganisms used in food fermentations**
- **Dairy Fermentations-starter cultures, types and methods of preservation and propagation, Lactic acid and aroma compounds production, Health benefits of LAB, probiotics, prebiotics and symbiotic**
- **Fermented Foods-types, methods of manufacture for vinegar, sauerkraut, tempeh, miso , soya sauce ,beer, wine and traditional Indian foods**

Unit III

Food borne Diseases

- **Types – food borne infections, food borne intoxications and toxic infections**
- **Origin, General characteristics , symptoms and prevention of some commonly occurring food borne diseases caused by Bacillus spp., by Clostridium botulinum, Staphylococcusaureus ; Clostridium botulinum; C. perfringen; Listeria monocytogene; Salmonella; Escherichia.coli; Yersinia enterocolitica; Shigella spp., Vibrio parahaemolyticus, Mycotoxins, Hepatatis, Gastroenteritis viruses, Entamoebahistolytica and Entamoeba coli..**
- **Emerging pathogens of concern**

Unit IV

- **Microbiology of Water and Food Commodities: Microbiology of water and their importance of processing of foods in industries. MPN of coliforms Microbiology of milk – Hetero and homo fermentative Lactic acid bacteria – Yogurt and Cheese fermenting**

Trends in Food Microbiology

- **Rapid Methods of Detection**
- **SCP and SCO**

Recent Advances

- **Microbial quality control: Determination of microorganisms in foods by cultural, microscopic, physical, chemical methods.**

Practical

Credits: 1

Practical: 40
Internal Assessment: 10
Total Marks: 50

1. **Introduction and study of microbiological instruments.**
2. **Media preparation, aseptic techniques and transfer of microorganisms.**
3. **To study various culture techniques- pour plating, spread plating and streaking.**
4. **To study morphology of bacteria by simple staining and negative staining.**
5. **Principle, procedure and use of gram staining method.**
6. **Staining of bacterial spores.**
7. **To distinguish the growth characteristics of microorganisms in various differential and selective media.**
8. **Identification of fungi by Lactophenol cotton blue staining method.**
9. **Sampling and observation of microorganisms from natural sources.**
10. **To study serial dilutions of the sample and plate counts.**
11. **Standard plate count for raw, pasteurized milk and ice-cream.**
12. **Presumptive coliform test for milk, butter and ice-cream.**

13 Suggested Resources for Reading

1. Tortora G.J et al, 2013, Microbiology: an introduction, Pearson Education
2. Nester, E.W, 2011, Microbiology, McGraw-Hill Higher Education
3. Dubey, R.C., Maheshwari, D.K., 2011, Textbook of Microbiology, S.Chand Publications, ND
4. Adams, M.R., Moss, M.O., 2012, Food Microbiology, New Age International Pvt. Ltd., ND.
5. Pelczar, Reid and Chan, 2008, Microbiology, McGraw hill Ed, ND
6. Brock biology of Microorganisms, 2011, Benjamin Cummings publishers.
7. Ingraham, John.L.2009, Introduction to Microbiology, 3 Ed., Thomson brocks/Cole Inc.
8. Ananthanarayan, Panikar, CKJ.,2013, Textbook of Microbiology, Oriental Longman Pvt. Ltd., Hyderabad.
9. Frazier, William, C. 2008, Food Microbiology, Tata McGraw Hill Ed., ND.

Note: Latest edition of text books may be used.

1	**Course Title**	**FOOD ANALYSIS AND INSTRUMENTATION**
2	**Course Code**	**302**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><th>External Theory</th><th>Internal Theory</th><th>External practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table> • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Food Microbiologist**
8	**Course Objectives**	**To generate the skill of handling the different instruments food analysis.**
9	**Course Outcomes**	On successful completion of the course, the Students will be able CO1: **To handle the different instruments of food process technology.** **CO2: To study the various techniques of food analysis.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**1. The syllabus of this paper has been divided into FOUR units.** **2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **4. All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

Unit 1

- **Introduction to food analysis, type of samples and sampling techniques, storage and preservation of samples, expression of results. Different preservation principles involved in food processing Sampling: Sampling techniques and preparation of food samples. Sampling of Milk, Eggs, Dried & Frozen food products.**
- **Physico-chemical methods for food analysis: Moisture & Total solids Carbohydrates Proteins Fats Fiber Ash & its types Minerals Vitamins Enzymatic methods**
- **Sensory Tests: Difference, Rating & Sensitivity tests. Types of panels Testing area & schedule**

Unit II

- **Instrumentation in food analysis : Principles, types and applications of colorimetry and electrophoresis and chromatography :paper, thin layer, ion exchange**

Unit III

- **Instrumentation in food analysis: Color measurement in foods. X-ray analysis of foods and its applications, scanning electron microscopy (SEM) in food analysis and identification**

Unit IV

- **Ultrasonic and other instruments for determination of physical and rheological properties of food Texture analysis in foods.**
- **Sensory versus instrumental analysis of texture, rapid methods of microbial analysis, immunoassays; Techniques for estimation and analysis of toxins and pesticides in food.**

Practical:

Credits: 1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. **Use of spectrophotometer in food content estimations.**
2. **Determination of physical and rheological properties of foods by different techniques.**
3. **Separation of food components by chromatography.**

13 Suggested Resources for Reading

1. **SemihOtles. 2008. Handbook of Food Analysis Instruments. CRC Press**
2. **Gulaboski, Rubin and Carlos M, Pereira (2008) Electrochemical techniques and instrumentation in Food Analysis in Handbook of Food Analysis Instruments (book, SemihOtles, ed.). In: Handbook of Food Analysis Instruments. Wiley.Practicals**
3. **Nielsen, S.S, 2004, Introduction to chemical Analysis of foods, CBS Publishers, New Delhi.**
4. **Ranganna. S., 2001, Handbook of Analysis & Quality control for Fruit & Vegetable Products, Tata McGraw Hill, New Delhi.**
5. **Pomeranz.Y, Meloan.C.E, 1996, Food Analysis – Theory & Practice, CBS Publiushers, New Delhi**
6. **Jacobs.M.B., 1999, Chemical Analysis of Food & Food Products, CBS Publiishers, New Delhi.**
7. **Jay.J.M, 1996, Modern Food Microbiology, CBS Publishers, New Delhi.**
8. **Debnath, 2005, Tools & Techniques of Biotechnology, Pointer Publishers, Jaipur**

Note: Latest edition of text books may be used.

<table>
<tr><td>1</td><td>Course Title</td><td>Microbiological Analysis</td></tr>
<tr><td>2</td><td>Course Code</td><td>303</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>45 hours of lectures for Theory + 30 hours of practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
Groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><th>External Theory</th><th>Internal Theory</th><th>External practical</th><th>Internal Practical</th></tr>
<tr><td>40</td><td>10</td><td>40</td><td>10</td></tr>
</table>

The exam of this course will be of 3hours duration.
There will be no objective type questions.
Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees

</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Microbiologist</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To understand the skill of handling the different instruments of microbiological analysis</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able
CO1: To handle the different instruments of microbiological analysis
CO2: To understand the various techniques of microbiological analysis</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>The syllabus of this paper has been divided into FOUR units.
Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
All questions carry equal marks.</td></tr>
</table>

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

Unit I

- **Equipments and tools used for of food products such as weighing scale, homogenizer, autoclave, laminar air flow chamber, vacuum pump, Bunsen burner, inoculation loop, incubator, refrigerator, microscope, etc.**
- **Microbiological techniques for analysis of food products**
- **Analyze results and provide recommendations SOPs for preparing culture media,**
- **Prepare liquid and solid culture media (nutrient broth and nutrient agar) following SOP and prepare for sterilization**

Unit II

- **Sterilization of culture media, solvents and glassware**
- **Disposal of used microbiological media and culture**
- **Quality standards for food products produced in the organization and methods to assess and maintain quality of food products**
- **GMP**

Unit III

- **Cleaning process to disinfect equipments/ tools and glass wares used ,Recommended sanitizers**
- **Organization standards to clean laminar air flow cabinet or lab bench using approved disinfectants and sanitizers, health and safety, recommended dosage for use of sanitizers, control of substances hazardous to health, handling/storage/ disposal/ cautions of use of sanitizers and disinfectants, fire precautions, occurrences, hygiene practice, disposal of waste, environmental protection, etc.**
- **Standards and procedures followed in the organization for cleaning and disinfecting equipment/ tools in microbiology laboratory**

Unit IV

- **Job responsibilities/duties to maintain lab hygiene standards.**
- **Dress code.**
- **Hygiene, safety and quality standards.**
- **Hygiene requirements and standards relevant to microbiology laboratory.**
- **Food Safety Standards and Regulations (as per FSSAI), food safety systems like HACCP, food hygiene and sanitation.**
- **Knowledge of legal regulations pertaining to work place- food regulatory system such as FSSAI and food standards for products produced in the organization.**
- **Microbiological hazards in production process and its critical control point to minimize or prevent those hazards.**
- **Physical, chemical and biological hazards and methods of prevention of various hazards**
- **Personal hygiene.**
- **Document and maintain records on microbiological analysis of raw materials for production of food products.**
- **Management and leading of team.**

Practical Credits: 1

Practical: 40
Internal Assessment: 10
Total Marks: 50

1. **Quality audit methods and procedures**
2. **lactophenol staining**
3. **Hanging drop preparation to observe motility of bacteria**

13 Suggested Resources for Reading

1. **Adams M.R and Moss M.O, 2004 "Food Microbiology", Panima Publishing corporation, New Delhi, 2ndEdition.**
2. **Frasier.1987. Food Microbiology.Tata McGraw-Hill Education Reference Books 1.Sivasankar B, 2009. "Food Processing and Preservation", PHI Learning Private Limited, Easterrn Economy Edition, 6th edition,**

3. **William C Frazier and Dennis C. Westoff , 2008. "Food Microbiology", Special Edition, Springer, TheMcGraw-Hill Companies,**
4. **Jay, James M. Modern Food Microbiology, CBS Publication, New Delhi, 2000**
5. **Garbutt, John. Essentials of Food Microbiology, Arnold, London, 1997**
6. **Pelczar MJ, Chan E.C.S and Krieg, Noel R. Microbiology, 5th Ed., TMH, New Delhi, 1993**

Note: Latest edition of text books may be used.

1	**Course Title**	**FOOD QUALITY TESTING AND EVALUATION**
2	**Course Code**	**401**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** (1 credit = 15 hours of theory learning based on 3 hrs./week)
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><th>External Theory</th><th>Internal Theory</th></tr><tr><td>80</td><td>20</td></tr></table> • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Food Microbiologist**
8	**Course Objectives**	• **To understand basic quality attributes of foods and their application in industry.**
9	**Course Outcomes**	On successful completion of the course, the Students will be able **CO1 To understand basic quality attributes of foods in raw as well as processed form.** **CO2 To understand the various systems of objective and subjective evaluation and their application in industry.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	• **The syllabus of this paper has been divided into FOUR units.** • **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** • **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** • **All questions carry equal marks.**

12 Course Content

Credits: 4
Theory: 80
Internal Assessment: 20
Total Marks: 100

UNIT: 1:

Introduction to quality attributes: Appearance, flavor, textural factors and additional quality factors.

Taste

- **Introduction**
- **organs involved in taste perception- tongue, papillae, taste buds, salivary glands**
- **mechanism of taste perception**
- **chemicals responsible for sweet, salt, sour, and bitter taste their structure and chemical dimensions**
- **Factors affecting taste quality, reaction time and factors affecting it absolute and recognition threshold, taste abnormalities**

UNIT II: Olfaction

- **Introduction and definition, anatomy of nose, mechanism of odour perception, Prerequisites for odour perception, odour classification, chemical specificity of odour, measurement of odour using different techniques – primitive, double tube olfactometer, Elseberg techniques, Wenzel's olfactometer, sniffing, merits and demerits of each methods, olfactory abnormalities.**

UNIT III: Colour

- **Introduction to natural and synthetic colours**
- **functions of colour in foods**
- **Optical aspect of colour,**
- **perception of colour,**
- **objective evaluation, colour measurement using different systems**
- **Qualitative and quantitative analysis of colour,**
- **Reflectance spectrophotometry and Colorimetry.**

UNIT IV: Texture

- **Introduction**
- **Definition and classification of texture profile**
- **Subjective evaluation, phases of oral processing**
- **Objective analysis, rheological methods of texture measurement including rheological models**
- **Measurement of texture in various food groups viz. cereals, dairy, fruits and vegetables, meat and meat products**

13 Suggested Resources for Reading

1. **Amerine,Pangborn&Roessler, Principles of sensory evaluation of food, Academic Press, London,1965**
2. **DeMan, 3rd edition,Principles of Food Chemistry, Springer,2007.**
3. **Meilgard, Sensory evaluation Techniques, 3rd ed CRC Press LLC, 1999**
4. **YeshajahuPomeranz& Clifton E. Meloan, Food Analysis & Theory & Practice,1st Indian ed. CBS Publisher & Distributors, New Delhi, 2002**

Note: Latest edition of text books may be used.

1	**Course Title**	**TECHNOLOGY OF CEREALS, PULSES AND OILSEEDS**
2	**Course Code**	**402**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of Practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><th>External Theory</th><th>Internal Theory</th><th>External practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table> • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Food Microbiologist**
8	**Course Objectives**	• **To create awareness about the processing of major cereals like paddy, maize etc.** • **To study the storage and handling techniques of cereals.** • **To gain knowledge on processing and milling of pulses.**
9	**Course Outcomes**	On successful completion of the course, the Students will be able • **CO1: To understand the technology of milling of various cereals** • **CO2: To understand the techniques of pulses and oilseeds refining** • **CO3: To understand the byproducts obtained during processing along with their uses.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	• **The syllabus of this paper has been divided into FOUR units.** • **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** • **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** • **All questions carry equal marks.**
12	**Course Content**	**Credits: 3** **Theory: 40** **Internal Assessment: 10** **Total Marks: 50**

UNIT 1: Technology of cereals

- **Wheat --Types, milling, flour grade, flour treatments (bleaching, maturing), flour for various purposes, technology of dough development. Chapter4-7, Kent**
- **Paddy Processing: Composition of paddy, physicochemical properties and Quality characteristics. Curing of Paddy – Parboiling Processes. Physico-chemical changes during parboiling – Cold Water soaking and Hot water soaking processes. Methods of grain drying- LSU Dryer, By Products of Paddy Processing - Paddy husk and its uses as husk ash, activated carbon, furfural and other byproducts. Production of Flattened and Puffed Rice from Paddy.**
- **Rice Milling: (mechanical & solvent extraction), Paddy Dehusking Processes. Rice Milling Flow Chart. Engelberg Huller Mills. Modern Rice Mills – their Components - Pre Cleaners, Shellers, Under Runner Shellers and Centrifugal Shellers. Paddy Separators – Satake and Schule Designs, Polishers -Cone polishers and other types, Bran and Brokens separators. Rice Mill yields and loss due to brokens at different stages of milling. Use of Rice Bran in Edible oil Industry.**

UNIT II:

- **Barley- Milling (pearl barley, barley flakes & flour), beer preparation**
- **Oats – Milling (oatmeal, oat flour & oat flakes)**
- **Sorghum and millets – Traditional & commercial milling (dry &wet)**

UNIT III:

- **Milling Of Pulses: Major Pulses grown in the country .Traditional milling process -merits and demerits. Dry milling, Wet milling & Improved milling method**
- **Drying of legumes - Sun and Traditional drying processes steps – Precleaning, Pitting, Oil application, conditioning, dehusking and splitting - Machinery and equipment employed. Modern milling process - Process flow chart -Mechanical hot air drying and conditioning - merits and demerits, dehusking in Pulse Pearler, Water conditioning, splitting of pulses in Pulse splitter, Merits and demerits. Mini dhal mill -working principle - advantages and disadvantages. Grinding of split pulses, pulse flour products, their applications and equipments used.**

UNIT IV:

- **Milling And Processing Of Maize: Products of milling - Flour, Semolina, Brewers' grits etc and their applications. Gluten and Starch Separation, Equipment needed for Degermination, Debraning and starch separation. Starch conversion into other value added products – Acid Hydrolysis, Enzyme Hydrolysis, Isomerization processes.Processing for Dextrose, Malto Dextrin and other products. Extraction and refining of Corn oil in brief. Grain storage and handling**

Technology of Oilseeds

- **Introduction**
- **Sources of protein (defatted flour, protein concentrates and isolates), properties and uses, protein texturization, fiber spinning**
- **Breakfast cereals – porridge , flakes and puffed products.**

Practical

Credits:1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. **Determination of physical properties of different cereal grains.**
2. **Determination of water absorption capacity, sedimentation value and alcoholic acidity of the Maida.**
3. **Determination of adulterant ($NaHCO_3$) in wheat flour/ Maida.**
4. **Estimation of Protein content of different Cereals and Legumes.**
5. **Determination of Gluten content in wheat flour samples.**
6. **Physical characteristics of wheat.**
7. **Moisture content of wheat and products.**
8. **Yeast fermenting power.**
9. **Physical characteristics of rice.**
10. **Cooking quality of rice.**

13 Suggested Resources for Reading

1. Chakraverty, A.: Post Harvest Technology of Cereals, Pulses and Oilseeds. Oxford and IBH Publishing Co, Calcutta, 1995.

2. Samuel A .Matz: "The Chemistry and Technology of Cereals as Food and Feed", Chapman and Hall, 1992. Reference Books 1.Smartt J, "Tropical Pulses", Longman Group Ltd. London, 1980.

3. Pomeranz Y, "Modern Cereal Science and Technology" VCH Publishing Inc. New York, 1987.

4. Cryde M. Christensen, "Storage of Cereal Grains and their Products", American Association of Cereal Chemists inc., St. Paul, USA, 1982. 5

5. Bernard Godon and Claude Willm, "Primary Processing of Cereals" Berns and Noble Publishers, 1994.6.KarelKulp and Joseph P Pante, "Handbook of Cereal Science and Technology", MercelDekkar, USA, 2000.

6.Kent, Technology of Cereal, 5th Ed. Pergamon Press, 2003

7. Marshall, Rice Science and Technology, Wadsworth Ed., Marcel Dekker, NewYork, 1994

Note: Latest edition of text books may be used.

1	**Course Title**	**FOOD SAFETY**
2	**Course Code**	**403**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 15 hours of Practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><th>External Theory</th><th>Internal Theory</th><th>External practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table> • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Food Microbiologist**
8	**Course Objectives**	• **To understand the concerns of Food safety and hygiene**
9	**Course Outcomes**	On successful completion of the course, the Students will be able • **CO1: To understand the types of hazards associated with food** • **CO2:To understand the food regulations (national as well as international)** • **CO3: To design and implementation of food safety management systems such as ISO series, HACCP and its prerequisites such as GMP, GHP etc.** • **CO4: To understand the emerging concerns of food safety and hygiene.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	• **The syllabus of this paper has been divided into FOUR units.** • **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** • **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** • **All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

UNIT I:

Introduction to Food Safety

- **Definition**
- **Factors affecting Food Safety**
- **Importance of Safe Foods**

Food Hazards of Physical and Chemical Origin

- **Introduction**
- **Physical Hazards with common examples**
- **Chemical Hazards(naturally occurring ,environmental and intentionally added)**
- **Impact on health**

Food Hazards of Biological Origin

- **Introduction**
- **Indicator Organisms**
- **Food borne pathogens: bacteria**
- **Food borne pathogens: viruses**
- **Food borne pathogens: eukaryotes**
- **Seafood and Shell fish poisoning**
- **Mycotoxins**

UNIT II:

Management of hazards

- **Need**
- **Control of parameters**
- **Temperature control**
- **Food storage**
- **Product design**

Hygiene and Sanitation in Food Service Establishments

- **Introduction**
- **Sources of contamination**
- **Control methods using physical and chemical agents**
- **Pest and Rodent Control**

Unit III

Food Safety Management Tools

- **Basic concept only**

Microbiological criteria

- **MRA**
- **Microbiological standards and limits (for processed food, water)**
- **Microbiological Assessment and categories of food based on microbial quality**
- **Sampling**
- **Basic steps in detection of food borne pathogens**
- **Water Analysis**

UNIT IV:

Food laws and Standards

- **Indian Food Regulatory Regime**
- **Global Scenario**
- **Other laws and standards related to food**

Recent concerns

- **New and Emerging Pathogens**
- **Packaging ,Product labeling and Nutritional labeling**
- **Genetically modified foods \ Transgenic**
- **Organic foods**
- **Newer approaches to food safety**
- **Recent Outbreaks**

Practical

Credits:1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. **Preparation of different types of media (complex, differential and selective)shift to microbiology**
2. **Microbiological Examination of different food samples**
3. **Bacteriological Analysis of Water**
4. **Assessment of surface sanitation by swab/rinse method**
5. **Assessment of personal hygiene**
6. **Scheme for the detection of food borne pathogens**
7. **Implementation of FSMS – HACCP, ISO : 22000**

13 Suggested Resources for Reading

1. **Lawley, R., Curtis L. and Davis,J. The Food Safety Hazard Guidebook , RSC publishing, 2004**
1. **De Vries. Food Safety and Toxicity, CRC, New York, 1997**
2. **Marriott, Norman G. Principles of Food Sanitation, AVI, New York, 1985**
3. **Forsythe, S J. Microbiology of Safe Food, Blackwell Science, Oxford, 2000 &Sons; USA, 1987**

Note: Latest edition of text books may be used.

<table>
<tr><td>1</td><td>Course Title</td><td>FOOD QUALITY MANAGEMENT</td></tr>
<tr><td>2</td><td>Course Code</td><td>404</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration/ LP
(Lectures)
(Practical)</td><td>45 hours of lectures for Theory + 15 hours of Practicals
(1 credit = 15 hours of theory learning based on 3 hrs./week)
(1 credit = 30 hours of practical based on 2 hrs./week)
(1 Semester = 15 weeks)
Groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td><table><tr><th>External Theory</th><th>Internal Theory</th><th>External practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table>
The exam of this course will be of 3hours duration.
There will be no objective type questions.
Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees</td></tr>
<tr><td>7</td><td>Job Role</td><td>Food Microbiologist</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>To learn about quality management in food production chain.
To learn about physical, chemical contaminants in foods
To learn about latest trends and techniques in food science</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the Students will be able
CO1: To understand the significance of safe processing of foods.
CO2: To understand the role of food standards and regulations in maintaining food quality.
CO3: To understand the different principles of food quality control.
CO4: To assess the food quality assurance of different food.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>The syllabus of this paper has been divided into FOUR units.
Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
All questions carry equal marks.</td></tr>
</table>

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

UNIT I

Food Quality

- **Introduction to food quality management – Definition, quality concepts, quality, quality perception, quality attributes, safety, health, sensory, shelf life, convenience, extrinsic attributes, factors affecting food behavior. Principles of food quality control and quality assurance Objectives, importance and functions of quality control. Quality management principles and methods, tools and techniques.**
- **Organization policies and goals, quality management, budget management, food regulatory policies and procedures related to products produced in the organization, quality mark accreditations of the organizations, audit procedures, code of business conduct, leadership techniques, manpower modeling and handling**

- **Quality in the Agri- food production chain-Techno- managerial approach, food quality relationship and food quality management functions. Dynamics on the agri-food production chain, core developments in food quality management.**

UNIT II

Food contamination

- **Contamination in Food- : Physical, chemical (heavy metals, pesticide residues, antibiotics, veterinary drug residues, dioxins, environmental pollutants, radionucleotides, solvent residues, chemicals) Natural toxins.**
- **Total quality management (TQM) - good manufacturing practices, good hygienic practices, good lab practices Microbial quality control : determination of microorganisms in foods by cultural, microscopic, physical, chemical, immunological and bioassay methods**

UNIT III

- **Food adulteration, nature of adulteration, methods of evaluation of food adulterants and toxic constituents of milk, tea, coffee, meat, spices, cereals, oils & fats and other products. Their analysis and analysis of common preservatives used in processed foods. Permissible limits of Preservatives.**
- **Sensory quality evaluation: Introduction, methods, panel screening, selection methods, Sensory and instrumental analysis in quality control**
- **Introduction, need of food additives in food processing and preservation. Characteristics and classification of food additives.**
- **Antimicrobial agents. -Nitrites, sulphides, sulphur di oxide, sodium chloride, hydrogen peroxide.**
- **Antioxidants - Introduction, mechanism of action, natural and synthetic anti-oxidants, technological aspect of antioxidants.**
- **Colors- Introduction, importance, classification- natural, artificial, and natural identical, FD&C Dyes and Lakes. Use of plant tissue culture, polymeric colors etc. for color**
- **Quality control of following food products : Milk & milk products Oils & Fats Cereal grains & flours Fruits & vegetable products Canned foods Egg & egg products Meat & Meat products 8.**

Unit IV

- **Food Laws, standards and regulations: History, National and International laws & Regulations: USFDA, EU, USFDA, ISO 9000 series etc. Food laws and standards, IPR and patents, Food standards and safety Act : salient provisions and prospects, role of various national and international agencies**
- **World Trade Organization (Sanitary and Phyto Sanitary agreement, Technical Barriers in Trade), -Standards of Identity, Standards of fill of the container. Statistical quality control in food industry Food safety and Standards Act 2006: Salient provision and prospects, Role of national and international regulatory agencies, Bureau of Indian Standards (BIS), AGMARK, Food Safety and Standards Authority of India (FSSAI**

Practical

Credits:1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. **Qualitative tests for fats and oils, spices and condiments.**
1. **Estimation of residual sulphur dioxide in beverages.**
2. **Chromatographic estimation of colour.**
3. **Analysis of edible common salt for MC, MIW and total chlorides.**
4. **Estimation of pesticide residues in food/water.**
5. **Estimation of benzoic acid in foods.**
6. **Techniques of quality assessment of different natural and processed foods.**
7. **Identification and ranking of food product attributes**
8. **Sensory and instrumental methods for measuring food quality assessment of fruits, vegetable, dairy products, milk and other processed products.**

13 Suggested Resources for Reading

1. **Pieternel A, Luning, Willem J. Marcelis, Food Quality Management, Technological and Managerial principles and practices, Wageningen,2009.**
2. **Brannen and etal,Food Additives, Marcel Dekker,New York,1990**
3. **Shalton, Principles and Practices for the safe processing of Foods.**
4. **DeMan, 3rd edition, Principles of Food Chemistry, Springer, 2007.**
5. **InteazAlli (2003). Food Quality Assurance Principles and Practices. CRC Press .**
6. **J Andres Vasconcellos (2003) Quality Assurance for the Food Industry. A Practical Approach.. CRC Press**

Note: Latest edition of text books may be used.

1	**Course Title**	**Principles of Management**
2	**Course Code**	**501**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** (1 credit = 15 hours of theory learning based on 3 hrs./week)
5	**Credits**	4 Credits
6	**Max. Marks and Time**	**External Theory: 80** / **Internal Theory: 20** • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Production Manager**
8	**Course Objectives**	• **To make the students understand the concepts of management & their Practical application in the food industry**
9	**Course Outcomes**	On successful completion of the course, the Students will be able • **CO: To understand the concepts of management & their Practical application in the food industry.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	• **The syllabus of this paper has been divided into FOUR units.** • **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** • **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** • **All questions carry equal marks.**
12	**Course Content**	

Credits: 4
Theory: 80
Internal Assessment: 20
Total Marks: 100

PRINCIPLES OF MANAGEMENT

UNIT I

Organization: Concept of organization, Goals and policies, Nature & importance of organization. Standards of business processes of organization.
Principles of organizations: Formal & Informal, Centralized & Decentralized. Resource management. Code of business conduct and manpower modeling.

Management: Concept of Management. Need and Scope. Levels of management. Types of management. External & internal factors affecting management. Different schools of management thought – Behavioral, Scientific, Systems and Contingency.

UNIT II

Planning: Definition, Nature and importance of planning. Advantages & disadvantages. Types of plans, objectives, strategies, policies, procedures, methods, rules, programs & budgets. Steps in planning. Planning assumptions.
Decision-making: Types of decisions. Step by step decision making process.
Coordination and Controlling: Coordination: Concept, need and techniques. Controlling Objectives and Process of control. Devices of control.

UNIT III

Communication: Definition, Nature, Process of communication.
Types of communication: Upward & downward, Verbal & Nonverbal, Formal & Informal.
Barriers to communication, Methods of improving communication and effectiveness.
Staffing Definition, Delegation and Departmentalization. Authority & responsibility. SWOT Analysis of staff. Recognizing and rewarding staff members.

UNIT IV

Leadership: Definition. Leadership theories: Behavioral, contingency and emerging leadership theories. Different styles of leadership. Characteristics of a good leader.
Motivation: Definition. Nature & importance. Benefits of motivated staff. Theories of motivation: Maslow's theory of need hierarchy, Herzberg's two factor theory, McGregor's theory 'X' and theory 'Y'.

13 Suggested Resources for Reading

Psychology for managers by Shashi K. Gupta and Rosy Joshi -Kalyani Publishers, 2007.
Principles of Management by T. Ramasamy -Himalaya Publishing house, 2010.
Principles of Management by Shikha Taneja -S K Kataria & Sons Publication, 2012.

Note: Latest edition of text books may be used.

1	**Course Title**	**Production Planning In Food Industry**
2	**Course Code**	**502**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 15 hours of Practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table> • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Production Manager**
8	**Course Objectives**	**To make students understand the production planning and subsequent production control.**
9	**Course Outcomes**	On successful completion of the course, the Students will be able **CO: To understand the production planning and subsequent production control.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	➢ **The syllabus of this paper has been divided into FOUR units.** ➢ **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** ➢ **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** ➢ **All questions carry equal marks.**

Course Content

Credits: 3
Theory: 40
Internal Assessment:10
Total Marks: 50

UNIT I

Production plan of food production and processing industry: Quality policy and objectives of food processing. Sales and customer needs/ market requirements.
Resource availability: Availability of raw materials and Packing materials. Equipment availability. Production capacity. Manpower requirement and availability, stock level, storage capacity, transport capacity.
Standard operating procedures: purchasing raw materials, receiving raw materials, storage, cleaning, holding, personal hygiene, facility and equipment.
Plant Layout: Overview and types of Plant Layout, Process Layout and Product Layout.

UNIT II

Maintenance Management: Concept, maintenance objectives. Identification of equipment requirements to meet production target. Equipment maintenance process and procedure, consumables and man power requirement. Analysis of equipment maintenance data. Linkage between production and maintenance personnel. Maintenance costs and cost minimization.

UNIT III

Schedule Production: Monitor production process. Identification of variation in production schedule. Adjustment of production schedule. Monitor production output and cost. Reschedule and Review production plan. Setting polices and plans.Initiatives to implement improvement opportunities for large production.

UNIT IV

Management of trials production: Objective of trial production and trial product, processing method and specification. Selection of production team for trial. Preparation of technical production procedures (considering all engineering and process parameters for new product trial).
Preparation of detailed trial production schedule. Monitoring trial production. Documentation and evaluation of trial production data and identification of process/parameters to be modified/ changed to achieve required specification.

Practical

Credit :1
Practical: 40
Internal Assessment: 10
Total Marks: 50

- **Prepare list of raw materials used for production and the finished dairy/ bakery products.**
- **Study of the process parameters and provision of necessary information to fill the process chart of any dairy/ bakery product.**
- **Prepare process charts for food product and processing of any dairy/ bakery product.**
- **Prepare report on equipment manuals, process documents and internal information to understand the equipment's operation and process requirement.**
- **Compile performance data of equipment to identify cause for lack of performance of any food processing industry.**

Suggested Resources for Reading

Economics and Management of the food industry by Jeffrey H. Dorfman- Routledge Publishers, 2013.
Energy Efficiency and management in food processing facilities by Lijun Wang- CRC Publications, 2008.

Note: Latest edition of text books may be used.

1	**Course Title**	**Production Optimization & Cost Efficiency**
2	**Course Code**	**503**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 15 hours of Practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table> • The exam of this course will be of 3 hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Production Manager**
8	**Course Objectives**	• To make students understand management of production optimization and cost efficiency by managing utilities and energy, optimizing production and implementing changes in production process**.**
9	**Course Outcomes**	On successful completion of the course, the Students will be able CO: To understand the management of production optimization and cost efficiency by managing utilities and energy, optimizing production and implementing changes in production process**.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	➢ **The syllabus of this paper has been divided into FOUR units.** ➢ **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** ➢ **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** ➢ **All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

UNIT I

Overview of production process: Analysis of equipment performance, process capability and change over time, maintenance, consumables, power.

Factors affecting performance of production and improvement opportunities.

UNIT II

Utilities and energy usage: Types of Production utilities in food industry. Estimating utilities and energy usage requirement. Energy conservation guidelines in food industry.

Procedures for promoting optimum utilities usage, Review of changes and implementation feedback. Methods for promoting renewable and efficient resource and utilities management.

UNIT III

Production process optimization: Identification of new food safety production methods. Designing new procedures, processes, systems and structures. Obligation with change management team/resource.

Methods for impact evaluation on product quality. Documentation of production process changes.

UNIT IV

Inventory Management: Objectives of Inventory Management. Importance of inventory control. Techniques of inventory control: Economic order quantity, ABC analysis and Perpetual inventory system.

Purchasing Management: Objectives and Functions of Purchasing. Purchasing cycle. Vendor Rating. Customer feedback.

Practical

Credit: 1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. **Preparation of report on different methods for optimum usage of energy in food industry in India.**
2. **Visit any food industry unit and prepare report on methods used for production process optimization.**
3. **Identify cost saving options and proposed changes in food production process.**
4. **Prepare flowcharts on purchasing cycle of different (Bakery/ Confectionary/ Dairy/ Fruit and vegetable) food processing unit.**
5. **Discuss laws introduced by Indian Government to promote efficient and renewable resources promotion in food industries.**

13 Suggested Resources for Reading

- **Energy Efficiency and management in food processing facilities by Lijun Wang-CRC Publications, 2008.**
- **Materials & Logistics Management by L.C. Jhamb -Everest Publishing House, 2005.**
- **Introduction to Materials Management by J. R. Tony Arnold, Stephen N. Chapman and Lloyd M. Clive-Pearson publications, 2010.**

Note: Latest edition of text books may be used.

1	**Course Title**	**Food Plant Layout and Business Ethics**
2	**Course Code**	**504**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** (1 credit = 15 hours of theory learning based on 3 hrs./week)
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><th>External Theory</th><th>Internal Theory</th></tr><tr><td>80</td><td>20</td></tr></table> - The exam of this course will be of 3hours duration. - There will be no objective type questions. - Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. - Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Production Manager**
8	**Course Objectives**	**Exposure of the students to the basic setup of a Food industry** **To make them conversant with the machinery and equipments in a food industry.** **The objective of this paper is to familiarize the students with the importance of ethics in business and understanding of issues related to corporate social responsibility and corporate governance.**
9	**Course Outcomes**	On successful completion of the course, the Students will be able **CO: To familiarize the students with the importance of ethics in business and understanding of issues related to corporate social responsibility and corporate governance.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	**The syllabus of this paper has been divided into FOUR units.** **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** **All questions carry equal marks.**

12 Course Content

Credits: 4
Theory: 80
Internal Assessment: 20
Total Marks: 100

Unit I

Basic concepts of industrial plant layout and design with special reference to food packaging industries. Application of HACCP concept, ISO, FPO & MPO requirements in food plant layout and design. Design consideration for location of food plants. Basic understanding of equipment layout and ventilation in food processing plants. Preparation of flow sheets for material movement and utility consumption in food plants.

Unit II

Plant layout and design of bakery and biscuit industries. Plant layout and design of fruits and vegetables processing industries including beverages. Plant layout and design of milk and milk products. Miscellaneous aspects of plant layout and design like provision for waste disposal, safety arrangements etc.

Unit III

Business Ethics: Meaning and Concept, Principles of Business Ethics, Characteristics of Ethical Organizations, Theories of Business Ethics. Globalization and Business Ethics, Stakeholder's Protection, Corporate Governance and Business Ethics. Ethical Issues in Indian Business.

Unit IV

Professional Values: Objectives, Morals, Values, Ethics, Integrity, Work ethics, Service learning, Virtues, Respect for others, Living peacefully, Caring, Sharing, Honesty, Courage, Valuing time, Cooperation, Commitment, Empathy, Self-confidence, Challenges in the work place, Spirituality. Corporate Social Responsibility: Social Responsibility of business with respect to different stakeholders, Arguments for and against social responsibility of business, Social Audit, Corporate Social Responsibility and Corporate Governance.

13 Suggested Resources for Reading

1. Manufacturing Facilities Design and Material Handling by Fred E. Meyers, and Matthew P. Stephens, 3rd Edition, Pearson Prentice Hall, 2000.
2. James M Moore, "Plant Layout and Design", McMillan & Co., (1959) 3. J M Apple, "Plant layout and Material handling", John Willey & Sons, (1977)
Practical based onFPP 205 Time: 3 hours 1. Industrial visit and report making.
3. J.P. Sharma, Corporate Governance, Business Ethics & CSR, Ane Books Pvt. Ltd., Nee Delhi.
4. Andrew Crane, Dirk Matten, Business Ethics, Oxford University Press, New Delhi.
5. Daniel Albuquerque, Business Ethics, Principles and Practices (Indian Edition), Oxford University Press, New Delhi
6. Fr. Floriano C. Roa, Business Ethics and Social Responsibility, Rexestore.
7. O. C. Ferrell, John Fraedrich, Linda Ferrell, Business Ethics: Ethical Decision Making & Cases, Cengagae Learning
8. Michael Blowfield, Alan Murray, Corporate Responsibility – A Critical Introduction, Oxford University Press, New Delhi

Note: Latest edition of text books may be used.

1	**Course Title**	**Food Biochemistry**
2	**Course Code**	**505**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** (1 credit = 15 hours of theory learning based on 3 hrs./week)
5	**Credits**	4 Credits
6	**Max. Marks and Time**	**External Theory: 80** / **Internal Theory: 20** • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Production Manager**
8	**Course Objectives**	The paper provides basic information on chemical, physical and functional properties of various biomolecules present in food and how they contribute to the overall quality of foods.
9	**Course Outcomes**	On successful completion of the course, the Students will be able CO : To understand the basic information on chemical, physical and functional properties of various biomolecules present in food and how they contribute to the overall quality of foods
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	➢ **The syllabus of this paper has been divided into FOUR units.** ➢ **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** ➢ **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** ➢ **All questions carry equal marks.**
12	**Course Content**	

Credits: 4
Theory: 80
Internal Assessment: 20
Total Marks: 100

UNIT I

Introduction to Bio molecules –

- Major and minor constituents of food, Bioavailability of nutrients, their functions, sources, Basics of bioenergetics.
- Carbohydrates - Classification, structure & properties, artificial sweeteners.
- Proteins - Structure and classification of amino acids, essential and non-essential amino acids, structural organization of proteins, physico-chemical properties of proteins, catabolism of proteins in prokaryotes.
- Lipids – Classification and its

UNIT II

- Biological membranes – Structure and composition in prokaryotes; Mechanisms of membrane transport systems in bacteria-Facilitated diffusion, Active and Passive transport and Group translocation.

UNIT III

- Enzymes- Enzymes classification and nomenclature; coenzymes, cofactor, kinetics, factors affecting catalytic efficiency of enzymes, allosteric enzymes, feedback inhibition; Competitive and noncompetitive inhibition.
- Vitamins and minerals - Classification, sources and functions.

UNIT IV

- Pigments – Introduction, major types and sources, Food pigments - Chlorophyll, carotenoids, anthocyanins, flavonoids, beet pigments, caramel.
- Flavours - Definition and basic taste factors, chemistry of food flavors from cocoa, coffee, vanilla beans and spices, flavor changes in fats and oil, flavor and aromas of food.
- Changes in food constituents during processing: Changes in carbohydrates on cooking, Browning and Maillard reactions, Effects of physic-chemical conditions on proteins, Denaturation of proteins, Oxidative and Hydrolytic Rancidity.

13 Suggested Resources for Reading

1. A L Lehninger, Nelson & Cox, (2013), Principles of Biochemistry, Worth Publishers, New York.
2. Bruce Alberts, Alexander Johnson, Julian Lewis, Martin Raff, Keith Roberts, Peter Walter, (2011), Molecular Biology of The Cell, Fifth Edition, Garland Science UK.
3. L Stryer, 2012, Biochemistry, W H Freeman and company, New York
4. Fennema, Owen,R.,ed, 2010, Food Chemistry
5. De, Amit Krishna., 2012, Biochemistry, S.Chand& Co. Ltd., New Delhi, India.
6. Nelson, David, L., 2011 ,Principles of Biochemistry, W.H.Freeman&Co., NY.
7. Rao, B.S., 2012, Experimental Biochemistry, IKI International Pvt. Ltd., ND.
8. Jain, J.L., 2010, Fundamentals of Biochemistry, S.Chand& Co. Ltd., ND.
9. Boyer, R. D., 2012, Modern Experimental Biochemistry, 3/e, Pearson Education Ptc. Ltd., ND.
10. Schaum's Outline of Theory & Problems of Biochemistry, 2011, Tata McGraw Hill, ed, ND.

Note: Latest edition of text books may be used.

1	**Course Title**	**Budgetary control and optimization**
2	**Course Code**	**601**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** (1 credit = 15 hours of theory learning based on 3 hrs./week)
5	**Credits**	4 Credits
6	**Max. Marks and Time**	**External Theory: 80** / **Internal Theory: 20** • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Production Manager**
8	**Course Objectives**	To make students understand the importance of budget in food industry and manage production within budget during production process in food processing unit.
9	**Course Outcomes**	On successful completion of the course, the Students will be able CO: To understand the importance of budget in food industry and manage production within budget during production process in food processing unit
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	➢ **The syllabus of this paper has been divided into FOUR units.** ➢ **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** ➢ **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** ➢ **All questions carry equal marks.**
12	**Course Content**	

Credits: 4
Theory: 80
Internal Assessment: 20
Total Marks: 100

UNIT I

Budget management: **Definition of Budget. Objectives, Types of budgets. Accounting models to manage budget. Budgetary systems. Monitoring methods. Evaluation of performance against budgets.**
General concept of Budget control: **Fundamentals of controlling activity. Principals of efficiency in the control process. Analysis of budget deviation.**

UNIT II

Production within optimum budget: **Allocation of budget for product/manpower**. **Scheduling production and planning timings. Identification of variances in budget control and revision. Ways of reducing expenditure.**

UNIT III

Variance analysis**: Types of variances: cost variances, material variances, labor variances, overhead and fixed overhead variances, sales variance, profit variance.**

Identification of variances: **Identify situations of actual budget exceeds the approved budget. Investigate reason for variance. Corrective measures to keep budget under control.**

UNIT IV

Identification of impact on budget of production-related decisions**: Scheduling holidays. Adjusting production volume. Scheduling equipment maintenance. Identification of opportunities to improve performance.**

Financial and accounting procedures of the organization: **Principles and processes involved in business and financial control.**

13 Suggested Resources for Reading

- **Purchasing and Inventory Management by K S Menon and Sarika Kulkarni -Shroff Publishers, 2011.**
- **Budgetary Control: A Management Tool for Measuring Performance by IfayemiOlayinka, Samuel Osundina -LAP Lambert Academic Publishing, 2012.**

Note: Latest edition of text books may be used.

1	**Course Title**	**DOCUMENTATION SYSTEM MANAGEMENT**
2	**Course Code**	**602**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table> • **The exam of this course will be of 3hours duration.** • **There will be no objective type questions.** • **Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.** • **Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees**
7	**Job Role**	**Production Manager**
8	**Course Objectives**	**To make students understand various documents in food industry and develop processes to pass audits.**
9	**Course Outcomes**	On successful completion of the course, the Students will be able **CO: To understand the various documents in food industry and develop processes to pass audits.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	➢ **The syllabus of this paper has been divided into FOUR units.** ➢ **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** ➢ **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** ➢ **All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

UNIT I

Documentation and record keeping in production plant: Importance of documentation. Training on documentation system. Up-to-date and accessible documents for audits on production process. Documentation of recommendation/corrective actions. Methods to track production information from documented and maintained records.
Documentation management strategies: Automation of accounts payable and receivable. Reduce Out-of-Spec Product costs by receiving electronic employee acknowledgments. Centralization of vendor contracts.

UNIT II

Standard Operating Procedures: Standard operating procedures: definition, procedure, purpose, format, developing, implementing and effective writing. Quality control policy and quality objectives of food processing company.
Food Safety Systems Manual: Food safety systems policies. Collection of documents with all validated and authorized written policies, procedures and programs.

UNIT III

Auditing Types, Audit requirement in production for food safety. Audit reports for different department- audit exercise.Audit procedures and audit requirements to ensure food safety, hygiene and sanitation in the organization. Methods of auditing to meet and maintain industry standards and regulatory requirements

UNIT IV

ERP: Definition. Evolution of ERP. ERP Characteristics and features. Role of ERP in food industry. Benefits of ERP. Business Process Reengineering (BPR). ERP implementation for online operations. Risk and governance issues in ERP. Co-relation of ERP with E-commerce.
Textbooks/References:

Practical

Credit :1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. **Prepare report on technical documents related to production process of the organization**
2. **Study legal and safety documents pertaining to food industry.**
3. **Prepare HACCP based SOP checklist of any food processing unit.**
4. **Report preparation of online ERP in the organization.**
5. **Discuss any 5-food safety policies.**

13 Suggested Resources for Reading

1. **Auditing in the food industry. From Safety and quality to environmental and other audits by Mike Dillon and Chris Grifith-CRC Press and Woodhead publishing limited, 2001.**
2. **Quality Assurance for the Food industry: A practical approach by Andres Vasconcellos J- CRC press, 2005.**

Note: Latest edition of text books may be used.

1	**Course Title**	**Food Industry Safety and Environmental Management**
2	**Course Code**	**603**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table> • **The exam of this course will be of 3hours duration.** • **There will be no objective type questions.** • **Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.** • **Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees**
7	**Job Role**	**Production Manager**
8	**Course Objectives**	**To ensure environmental safety and implement health and safety policies in food industry.**
9	**Course Outcomes**	On successful completion of the course, the Students will be able CO: To understand the environmental safety and implement health and safety policies in food industry.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	➢ **The syllabus of this paper has been divided into FOUR units.** ➢ **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** ➢ **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** ➢ **All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

UNIT- I

Safe work procedures: In production area and during production process for food industry.
Identification of safety and environmental hazards relevant to production processes. Environmental management system and environmental standards.

UNIT - II

Risk: Indicators, analysis and management. Causes of major safety failure: clothing and personal hygiene.
Sources of contamination: **Food safety testing methods. Basic concept of food safety regulations, guidelines and codes as per FSSAI. Hygiene and sanitation in food processing unit. Legal Basis for risk assessment. Role of material handling in food safety.**

UNIT - III

Procedures for disposal of waste. Waste management in production area and process. Methods of waste collection. Recycling or disposal for meeting industry requirements and environmental regulations. Basic concept of aerobic and anaerobic treatment.

UNIT- IV

Health and safety policies. Methods to establish systems for monitoring, measuring and reporting on health and safety. OHSAS – Occupational Health and Safety Management Systems.
Food regulatory systems**: Introduction to FSSAI, GMP, GHP, HACCP, QMS, ISO.**

Practical

Credit :1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. **Write project reports on waste management of food industry (Bakery/ Dairy/ Beverage/ Fruit and vegetable industry).**
2. **Read legal and safety, environmental and regulatory documents pertaining to the organization and prepare a report.**
3. **Prepare flowcharts for treatment of various food hazards in food industry.**
4. **Prepare flowcharts on various environment hazards in food industry.**
5. **Study of procedures for ensuring the quality and hygiene of the product compliance with FSSAI standards.**
6. **Prepare report on integrated management system as per HACCP & ISO.**

13 Suggested Resources for Reading

1. Safety, Health and Environmental concepts for the Process Industry by Michael Speegle -Delmar Cengage Learning, 2012.
2. Lewis' Dictionary of Occupational and Environmental Safety and Health by Jeffrey Wayne Vincol -CRC Press, 1999.
3. Waste Management in the Food industry by Gina Cybulska-Campden & Chorleywood-Food Research Association Group, 2000.

Note: Latest edition of text books may be used.

1	**Course Title**	**Food Packaging**
2	**Course Code**	**604**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **3-groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table> • **The exam of this course will be of 3hours duration.** • **There will be no objective type questions.** • **Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.** • **Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees**
7	**Job Role**	**Production Manager**
8	**Course Objectives**	To enable the students to understand about packaging and packaging materials, interaction of food items with packaging materials
9	**Course Outcomes**	On successful completion of the course, the Students will be able CO: To understand about packaging and packaging materials, interaction of food items with packaging materials
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	➢ **The syllabus of this paper has been divided into FOUR units.** ➢ **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** ➢ **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** ➢ **All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

UNIT I

Introduction:
• Classify packaging, Explain the packaging process and importance of packaging.
• Roles and responsibilities of a food products packaging technician
• Equipment used for food packaging, Tools and machineries used for food packaging, Identify and set the machines and tools required for production in working condition
• Effect of environmental factors on quality of food.
• General Approach, analysis of storage requirement, accelerated storage studies: Vacuum and Inert Gas Packaging: Tests on packaging materials, Mechanical strength (Tension, notch and tearing strengths), Gas and water vapour transmission rates.
• Packaging Methods – Form fill seal packaging , Hermetic closures, retortable pouches, Aseptic packaging Inert gas packaging , Active & intelligent packaging .

Unit II

• Packaging Materials – Properties of packaging materials in relation to their function (paper, glass, jute, wood, metal containers, flexible packaging materials, laminates, tetra packs). Quality control tests on packaging materials
• Packaging Criteria – Appearance, Protection, Function and cost.
• Estimation of shelf life.
• Packaging of food products – Factors determining the packaging requirements of various foods and Description of packaging of – a. Frozen products b. Dried products
c. Chemically preserved foods d. Fats and oils e. Confectionery f. Fruit juices g. Heat processed foods h. Fresh produce (Eggs, Fruits and Vegetables)
• Metal Cans As Packaging: Metallic can types - Tin cans and Aluminum cans. Specialty of Open top sanitary cans, Lacquers and their use, Three piece cans and Two piece cans, Aerosol Cans. Basics of Canning operations – Can Reformer, Flanger, Seaming,
Can closures. Glass jars and Bottles in food packaging, Design features and applications, Sterilization of bottles.

Unit III

• Flexible Films Packaging: Formation of Films and pouches, Plastics used and their Specific applications.
• Copolymers their applications.
• Co-extruded films and Laminates.
• Rigid and Semi rigid plastic packaging methods.
• Extrusion –Retort pouch packaging. Laminated Paper board Cartons, Fibre Board and Corrugated Card Board packaging and their applications.

Unit IV

• Filling and Sealing Operations For Various Types Of Packages: Closing and sealing of Rigid plastic containers.
• Filling and sealing of Flexible plastic containers, Seal types, hot wire sealing, hot bar sealing and impulse sealing. Form fill
• Seal equipment: Printing on packages, Bar codes, Nutrition labeling and legislative requirements. Filling and Sealing of pouches, pouch from fill seal machines.
• Active packaging, Moisture control, CO2 and Oxygen scavenging, modified atmosphere packaging –principles, applications.
• Permeability of gases in packs.
• Specialty packages.
• Personal hygiene and sanitation guidelines; • Food safety hygiene standards in a work environment, HACCP principles to eliminate food safety hazards in the process and products
• Materials and equipment used in cleaning and maintenance of work area and machinery,
• Methods of cleaning processes for tools and machineries used in packaging, Cleanliness of the process machineries required for production using recommended sanitizers
• Method of documenting and recording the details of raw material to final finished product
• Plan, Schedule and organize the work order to prevent potential problems

Practical

Credit :1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. Testing of Packaging material
2. To study the different packaging materials.
3. Determination of water vapour transmission rate for various packaging materials.
4. To determine grease resistance of packaging material.
5. To determine the wax content in given wax paper.
6. To estimate the basis weight of given packaging material.
7. To determine the chemical resistance of given packaging material.

13 Suggested Resources for Reading

- **Desrosier NW and Desrosier JN, The Technology of Food Preservation, CBS**
- **Publication, New Delhi, 1998**
- **Paine FA and Paine HY, Handbook of Food Packaging, Thomson Press India Pvt Ltd,**
- **New Delhi- 1992**
- **Potter NH, Food Science, CBS Publication, New Delhi, 1998**
- **Ramaswamy H and Marcott M, Food Processing Principles and Applications CRC**
- **Press, 2006**
- **Rao PG, Fundamentals of Food Engineering, PHI Learning Pvt Ltd, New Delhi, 2010**
- **Toledo Romeo T, Fundamentals of Food Process Engineering, Aspen Publishers, 1999**
- **Jood, Sudesh, 2002, Food Preservation , Agrotech Publisher Academy, Udaipur .**
- **Potter, N.N.,2002, Food Science, CBS Publishers,**
- **ND. Sethi, Mohini , 2001 Food Science, CBS Publishers,**
- **ND Srilakshmi, B., 2001, Food Science, New Age International Pvt. Ltd.,**
- **ND. Mahendru , S.N., 2000, Food Additives, Tata McGraw Hills, ND Manay, N.S., 2001, Foods : Facts & Principles , Wiley Eastern Ltd., ND.**
- **Coles, R., Dowell, D.M., Kirwan, J. "Food Packaging Technology". Black Well**
- **Publishing Ltd.ISBN-084939788X, 2003.**
- **NIIR Board: Food Packaging Technology Handbook National Institute of Industrial**
- **Research, New Delhi (2004)**
- **Pirenger O.G.and A.L.Baver: Plastic Packaging Materials for Food Wiley VCH, GmbH,**
- **Germany (2000).**

Note: Latest edition of text books may be used.

1	**Course Title**	**Applied Nutrition**
2	**Course Code**	**605**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**60 hours of lectures for Theory** (1 credit = 15 hours of theory learning based on 3 hrs./week)
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><th>External Theory</th><th>Internal Theory</th></tr><tr><td>80</td><td>20</td></tr></table> • The exam of this course will be of 3hours duration. • There will be no objective type questions. • Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. • Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	**Production Manager**
8	**Course Objectives**	• **Understand the basic concepts and principles of nutrition and health.** • **Gain knowledge about macronutrients and micronutrients, their functions, sources, and requirements.** • **Understand the role of food and nutrients in maintaining normal health and preventing malnutrition.** • **Learn about energy requirements and nutrient needs of the human body.** • **Develop the ability to relate food choices with nutritional adequacy and health outcomes.**
9	**Course Outcomes**	On successful completion of the course, the Students will be able to • **CO1: Define basic nutrition and health concepts and explain different forms of malnutrition.** • **CO2: Classify nutrients and describe their functions and dietary sources.** • **CO3: Explain food groups and relate them to a balanced diet.** • **CO4: Understand energy metabolism, BMR, and nutrient requirements (RDA).** • **CO5: Apply basic nutrition knowledge to promote healthy food habits and nutritional well-being**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.
11	**Instructions for paper setters and candidates**	➢ **The syllabus of this paper has been divided into FOUR units.** ➢ **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** ➢ **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** ➢ **All questions carry equal marks.**

12 Course Content

Credits: 4
Theory: 80
Internal Assessment: 20
Total Marks: 100

Unit I: Basics of Nutrition

Introduction to Nutrition

- **Meaning and definition of nutrition and health**
- **Relationship between nutrition and health**
- **Forms of malnutrition:**
  - **Under-nutrition**
  - **Over-nutrition**
- **Factors affecting food intake and nutritional status (economic, social, cultural, physiological)**

Nutrients

- **Definition of nutrients**
- **Classification of nutrients:**
  - **Macronutrients and Micronutrients**
  - **Organic and Inorganic nutrients**
  - **Energy-yielding and Non-energy yielding nutrients**
- **Importance of nutrients in the human body**

Unit II: Food and Carbohydrates **(Food-based understanding)**

Food

- **Definition of food**
- **Classification of foods**
- **Basic food groups (Indian food guide)**
- **Origin of foods: Plant and Animal sources**

Carbohydrates

- **Introduction and basic structure**
- **Functions of carbohydrates**
- **Dietary sources of carbohydrates**
- **Recommended Dietary Allowance (RDA)**

Unit III: Energy and Proteins **(Metabolism & body functions)**

Energy

- **Concept of energy in nutrition**
- **Basal Metabolic Rate (BMR)**
- **Factors affecting BMR**

Proteins

- **Basic composition of proteins**
- **Functions of proteins**
- **Dietary sources of proteins**
- **Recommended Dietary Allowance (RDA)**

Unit IV: Fats, Vitamins and Minerals **(Protective nutrients & health maintenance)**

Fats

- **Basic structure and types of fats**
- **Functions of fats**
- **Dietary sources**
- **Recommended Dietary Allowance (RDA)**

Vitamins

- **Classification:**
  - **Fat-soluble vitamins**
  - **Water-soluble vitamins**
- **General functions of vitamins**
- **Dietary sources**
- **Recommended Dietary Allowance (RDA)**

Minerals

- **Classification:**
  - **Major minerals**
  - **Trace (minor) minerals**
- **Functions of minerals**
- **Dietary sources**
- **Recommended Dietary Allowance (RDA)**

13 **Suggested Resources for Reading**

- **Textbook of Nutrition and Dietetics – Dr. Swaminathan**
- **Nutrition: Science and Applications – Lori A. Smolin & Mary Grosvenor**
- **Essentials of Food and Nutrition – Anita Tull &** others
- **Foods, Nutrition & Diet Therapy – Sunitha R. Nair**
- **Human Nutrition: Science for Healthy Living – Wendy J. Schiff**
- **Modern Nutrition in Health and Disease – A. Catherine Ross**
- **Normal and Therapeutic Nutrition – Susan A. Lanham-New**
- **Nutrition Through Life Cycle – Judith E. Brown**
- **WHO / FAO nutrition guidelines online**
- **ICMR nutrient requirement reports**
- **Indian Council of Medical Research (ICMR) dietary guidelines**

Note: Latest edition of text books may be used.

1	**Course Title**	**Food Waste Management and Sustainability**
2	**Course Code**	**606**
3	**Typology of Course**	**Core Discipline**
4	**Course Duration/ LP** (Lectures) (Practical)	**45 hours of lectures for Theory + 30 hours of practicals** (1 credit = 15 hours of theory learning based on 3 hrs./week) (1 credit = 30 hours of practical based on 2 hrs./week) **(1 Semester = 15 weeks)** **Groups will be created for the purpose of Practical for each unit**
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><th>External Theory</th><th>Internal Theory</th><th>External Practical</th><th>Internal Practical</th></tr><tr><td>40</td><td>10</td><td>40</td><td>10</td></tr></table> • **The exam of this course will be of 3hours duration.** • **There will be no objective type questions.** • **Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.** • **Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees**
7	**Job Role**	**Production Manager**
8	**Course Objectives**	1. **Introduce students to the concept, magnitude, and causes of food waste at global and national levels.** 2. **Develop an understanding of the environmental, economic, and social impacts of food loss and waste.** 3. **Familiarize students with food waste reduction strategies across the food supply chain.** 4. **Provide knowledge on utilization, valorisation, and energy recovery from food waste.** 5. **Promote awareness of sustainable food systems and responsible consumption practices.**
9	**Course Outcomes**	On successful completion of the course, the Students will be able to • **CO1: Define and classify food waste and explain its causes across the food supply chain.** • **CO2: Describe the environmental and economic impacts of food waste.** • **CO3: Identify strategies and technologies for reducing food waste at farm, industry, and consumer levels.** • **CO4: Explain methods of food waste utilization, valorisation, and waste-to-energy conversion.** • **CO5: Understand sustainability concepts and apply them to food systems and waste management practices.**
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions, Case Studies etc.

11	**Instructions for paper setters and candidates**	➢ **The syllabus of this paper has been divided into FOUR units.** ➢ **Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.** ➢ **The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.** ➢ **All questions carry equal marks.**

12 Course Content

Credits: 3
Theory: 40
Internal Assessment: 10
Total Marks: 50

Unit I: Introduction to Food Waste **(Concepts, causes, and impacts)**

- **Definition and classification of food waste and food loss**
- **Global and Indian scenario of food loss and food waste**
- **Causes of food waste along the food supply chain:**
  - **Farm level**
  - **Post-harvest handling and storage**
  - **Processing and distribution**
  - **Retail, food service, and household level**
- **Environmental impacts of food waste:**
  - **Land and water resource use**
  - **Greenhouse gas emissions**
  - **Economic and social implications**
- **Food waste quantification and auditing methods (basic concepts)**

Unit II: Food Waste Reduction Strategies **(Prevention and control measures)**

- **Farm-level food waste reduction technologies**
- **Improved post-harvest handling practices:**
  - **Controlled atmosphere storage**
  - **Cold chain management**
- **Industrial food waste minimization:**
  - **Lean manufacturing**
  - **Good Manufacturing Practices (GMP)**
  - **By-product recovery and utilization**
- **Role of stakeholders in food waste reduction:**
  - **Government policies and initiatives (FSSAI, MoFPI)**
  - **NGOs, food banks, and community-based programs**
  - **Consumer awareness and behaviour change strategies**

Unit III: Food Waste Utilization and Valorisation **(Resource recovery and circular economy)**

- **Concept of food waste valorisation**
- **Circular economy approach in food systems**
- **Conversion of food waste into value-added products:**
  - **Animal feed**
  - **Bio-fertilizers**
  - **Enzymes and organic acids**
  - **Bioactive compounds (polyphenols, pigments)**
- **Waste-to-energy approaches:**
  - **Biogas production**

- **Bioethanol**
- **Composting and vermicomposting**

- **Introduction to zero-waste and near-zero-waste food processing systems**

Unit IV: Sustainability in Food Systems **(Sustainable practices and future perspectives)**

- **Principles of sustainability and sustainable development**
- **Sustainable Development Goals (SDGs) related to food and waste**
- **Life Cycle Assessment (LCA) of food systems (basic approach)**
- **Sustainable food packaging:**
  - **Biodegradable and eco-friendly materials**
  - **Edible films and coatings**
- **Water and energy conservation in food industries**
- **Sustainable supply chains and green logistics**
- **Future trends in sustainable food systems:**
  - **Upcycled foods**
  - **Smart and active packaging**
  - **Regenerative agriculture**

Practical

Credit :1
Practical: 40
Internal Assessment: 10
Total Marks: 50

1. **Conduct food waste quantification in a kitchen/canteen/processing unit.**
2. **Demonstration of aerobic composting process using fruit & vegetable waste.**
3. **Development of a Value-Added Product from Food By-Product (For example: Okara-based snacks, Grape pomace powder blends)**
4. **Conduct a simple Life Cycle Assessment for a food product (e.g., bread, milk, rice).**

13 **Suggested Resources for Reading**

- **Food Waste Management: Principles and Practices – R. K. Soni & Neha Singh**
- **Food Waste Reduction and Valorisation – Maria Teresa Scarpato**
- **Sustainable Food Systems: From Agriculture to Industry – Kevin Hall**
- **Introduction to Food Engineering – R. Paul Singh & Dennis Heldman**
- **Food Processing Waste Management: Treatment and Utilization Technology – N. N. Potter**
- **Food Laws and Standards in India – FSSAI Publications**
- **Sustainable Development and Environmental Management – R. K. Sapra**
- **Waste Management and Resource Efficiency – World Bank Publications**
- **Life Cycle Assessment of Food Systems – R. Horne**
- **Handbook of Sustainable Food Processing – Charis Galanakis**

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